Matoshri College of Engineering and Research Centre

Department of Engineering Sciences

Presentation on

“Pressure Sensor”

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INTRODUCTION

- **Pressure Sensor**
- A **pressure sensor** is a device used to **measure the pressure of gases or liquids** and convert it into an **electrical signal** that can be monitored or controlled. Pressure sensors are widely used in **industrial automation, automotive systems, medical devices, and environmental monitoring**.
- These sensors detect pressure by sensing the **force applied on a sensing element** such as a diaphragm. The change in pressure causes a **mechanical deformation**, which is then converted into an **electrical output signal**.
- Pressure sensors help in **monitoring, controlling, and ensuring safety** in various systems like **water supply systems, air compressors, HVAC systems, and weather monitoring equipment**.

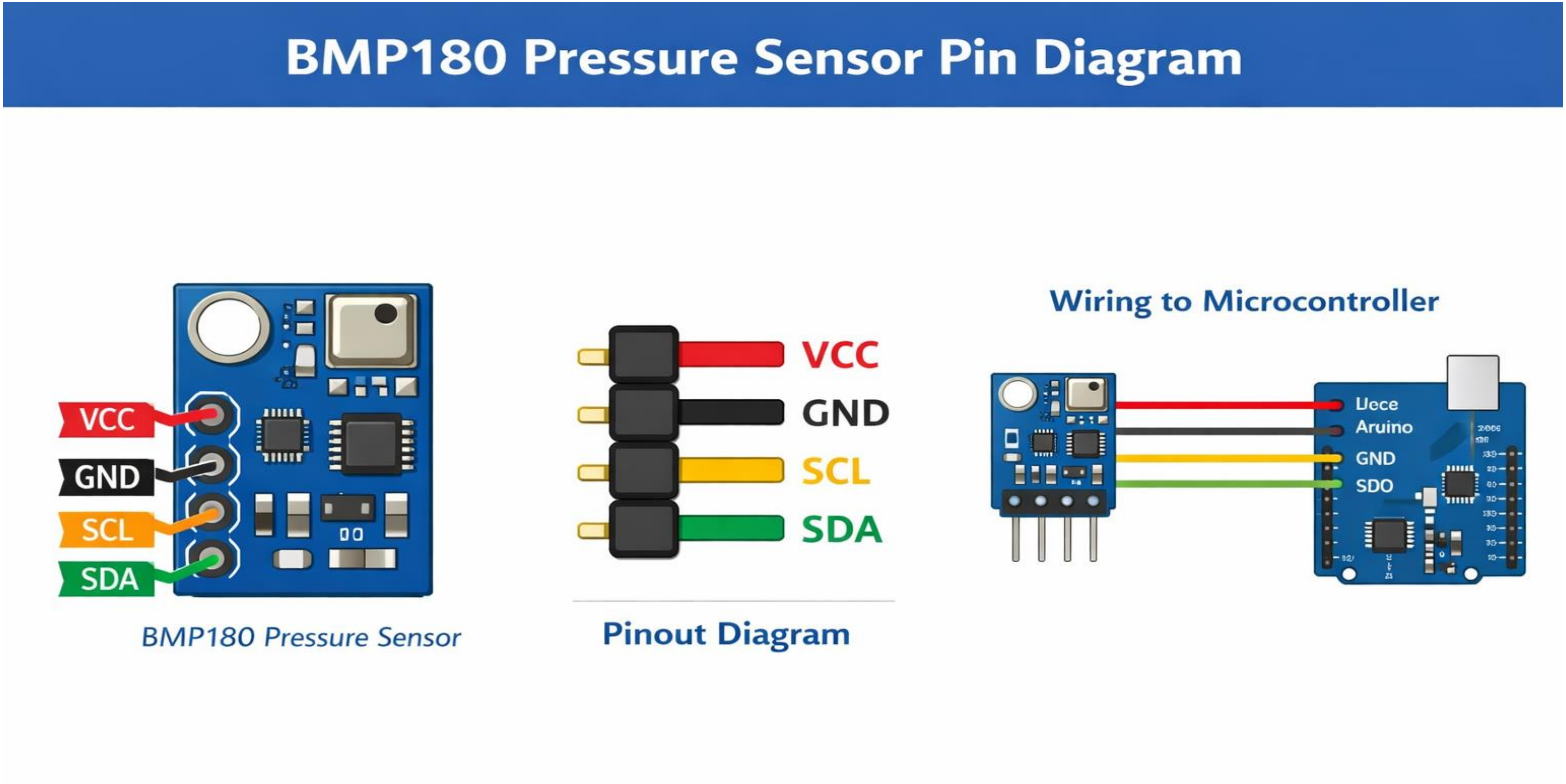
Sensor Type

- **Sensor Type**
- Pressure sensors are available in different types based on their working principle:
- **Piezoresistive Pressure Sensor** – Measures pressure using change in electrical resistance.
- **Capacitive Pressure Sensor** – Detects pressure by change in capacitance between plates.
- **Piezoelectric Pressure Sensor** – Generates electrical charge when pressure is applied.
- **Strain Gauge Pressure Sensor** – Uses strain gauges to measure deformation caused by pressure.

MANUFACTURER

- **Manufacturer**
- Some well-known manufacturers of pressure sensors include:
- Honeywell International
- Bosch
- Siemens
- TE Connectivity
- NXP Semiconductors
- These companies manufacture pressure sensors for **industrial, automotive, medical, and consumer electronics applications**.

PIN DIAGRAM



BMP180 Pressure Sensor Pin Diagram
VCC
GND
SCL
SDA
BMP180 Pressure Sensor
VCC
GND
SCL
SDA
Pinout Diagram
Wiring to Microcontroller
Uece
Aruino
GND
SDO

PIN DISCRIPTION

PIN NAME	FUNCTION
VCC	Power supply to the sensor (usually 3.3V or 5V).
GND	Ground connection.
SCL	Serial Clock Line used for I²C communication with a microcontroller.
SDA	Serial Data Line used for I²C communication to send sensor data.

ELECTRICAL CHARACTERISTICS

PARAMETER	VALUE
Operating Voltage	1.8 V – 3.6 V
Current Consumption	~5 µA (standard measurement)
Interface	I²C Digital Interface
Pressure Range	300 hPa – 1100 hPa
Operating Temperature	−40°C to +85°C
Resolution	Up to 0.02 hPa
Accuracy	±1 hPa

MECHANICAL DETAILS

PARAMETER	DETAILS
Package Type	LGA (Land Grid Array) package
Dimensions	Approximately 3.6 mm × 3.8 mm × 0.93 mm
Weight	Very lightweight (few grams when on module board)
Mounting Type	Surface Mount Technology (SMT)
Housing Material	Silicon sensor with metal lid
Port Type	Atmospheric pressure sensing port

PERFORMANCE SPECIFICATIONS

PARAMETER	SPECIFICATION
Pressure Measurement Range	300 hPa – 1100 hPa
Pressure Accuracy	±1 hPa
Pressure Resolution	Up to 0.02 hPa
Temperature Measurement Range	−40°C to +85°C
Temperature Accuracy	±2°C
Response Time	Very fast (few milliseconds)
Noise Level	Very low noise output

List of Application

- Pressure sensors are used in many fields such as:

1. **Weather Monitoring Systems**
2. **Altitude Measurement (Altimeters)**
3. **Smartphones and GPS Devices**
4. **Industrial Pressure Monitoring**
5. **Medical Equipment**
6. **Automotive Systems**
7. **HVAC Systems (Heating, Ventilation, Air Conditioning)**

Application
Weather Monitoring System

- Pressure sensors play an important role in **weather forecasting systems**. They measure **atmospheric pressure**, which helps predict changes in weather conditions.
- When atmospheric pressure **drops**, it usually indicates **rainy or stormy weather**. When pressure **increases**, it usually indicates **clear and stable weather**. The pressure sensor continuously measures the air pressure and sends the data to a **microcontroller or computer system**.
- Meteorological stations use this data to **analyze weather patterns and make accurate forecasts**. Because sensors like BMP180 are **small, low-power, and highly accurate**, they are widely used in **portable weather stations and environmental monitoring devices**.

CONCLUSION

- Pressure sensors are devices used to measure the pressure of gases or liquids and convert it into an electrical signal. They help in monitoring and controlling different systems to ensure safety and efficiency.
- These sensors are widely used in many fields such as industrial automation, automotive systems, medical equipment, and weather monitoring. Sensors like the **BMP180 Pressure Sensor** provide accurate and reliable pressure measurements.
- In conclusion, pressure sensors play an important role in modern technology. Their high accuracy, small size, and efficiency make them essential components in many electronic and automated systems.

REFERENCES

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5. Electronics Tutorials – Sensor Technology and Applications

Thank You